E-Commerce Order Management System

Task II: Identify Entities, Attributes, Primary Keys, and Relationships

1. Introduction

The E-Commerce Order Management System is designed to manage customer orders, products, suppliers, payments, shipments, and customer reviews. To design an efficient database, it is necessary to identify the entities involved in the system, define their attributes, determine the primary and foreign keys, and establish the relationships among them. This document presents the database entities and their relationships required for the proposed system.

2. Entity Identification

The following entities have been identified for the E-Commerce Order Management System:

- Customer
- Product
- Category
- Supplier
- Orders
- Order Details
- Payment
- Shipment
- Review

3. Entity Description

3.1 Customer

The **Customer** entity stores the personal and contact information of customers who purchase products from the e-commerce platform.

Customer Attributes

Attribute Name	Key Type	Description
Customer ID	Primary Key (PK)	Unique identifier of the customer
First_Name	-	Customer's first name
Last_Name	-	Customer's last name
Email	-	Customer's email address
Phone	-	Contact number
Address	-	Residential address
City	-	Customer's city
State	-	Customer's state
Postal_Code	-	Postal or ZIP code

3.2 Product

The **Product** entity stores all products available for sale in the e-commerce system.

Product Attributes

Attribute Name	Key Type	Description
Product_ID	Primary Key (PK)	Unique identifier of the product
Product_Name	-	Name of the product
Description	-	Product details
Price	-	Selling price
Stock_Quantity	-	Available stock quantity
Category_ID	Foreign Key (FK)	References Category
Supplier_ID	Foreign Key (FK)	References Supplier

Continue in the same format for:

- 3.3 Category
- 3.4 Supplier
- 3.5 Orders
- 3.6 Order Details
- 3.7 Payment
- 3.8 Shipment
- 3.9 Review

Each should have:

1. A **2–3 sentence description**.
2. A table of attributes.

4. Entity Relationships

The entities identified above are related to each other to ensure efficient management of the e-commerce system. The following table describes the relationships among the entities.

Source Entity	Relationship	Target Entity	Cardinality
Customer	Places	Orders	One-to-Many (1:M)
Orders	Contains	Order Details	One-to-Many (1:M)
Product	Appears In	Order Details	One-to-Many (1:M)
Category	Contains	Product	One-to-Many (1:M)
Supplier	Supplies	Product	One-to-Many (1:M)
Orders	Has	Payment	One-to-One (1:1)
Orders	Has	Shipment	One-to-One (1:1)

Customer	Writes	Review	One-to-Many (1:M)
Product	Receives	Review	One-to-Many (1:M)

5. Conclusion

The E-Commerce Order Management System consists of nine core entities with clearly defined attributes, primary keys, and foreign keys. The relationships between these entities ensure data integrity and support efficient order processing, inventory management, payment handling, shipment tracking, and customer feedback. This database design provides a strong foundation for implementing reliable and scalable e-commerce applications.